$\boxed{\text{Feb 19 2026}}$

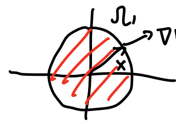
$\nabla^T(x) \cdot f(x) \le 0$

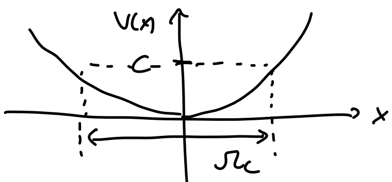
$\Omega_c := \{x : V(x) \le c\}$ $\quad c > 0$

↑

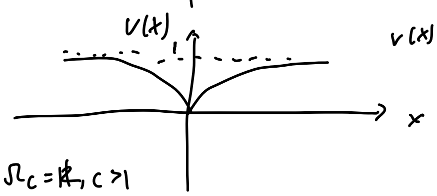
Level set of the Lyapunov set

Eg. $V(x) = x^2$





$V(x) = x^2$



$V(x) = 1 - e^{-x^2}$

$\Omega_c = \mathbb{R}, c > 1$

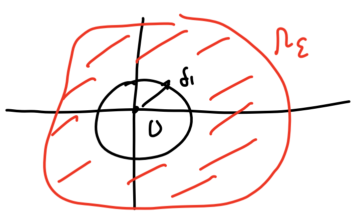
<u>Recall</u> $V(x)$ is continuous at $x = 0$

$\forall \varepsilon > 0, \exists \delta(\varepsilon) > 0$ s.t.

$|x - 0| \le \delta \Rightarrow |V(x) - V(0)| \le \varepsilon$

i.e. $x \in B_\delta \Rightarrow V(x) \le \varepsilon$

Ω_c is bounded for sufficiently small $c > 0$



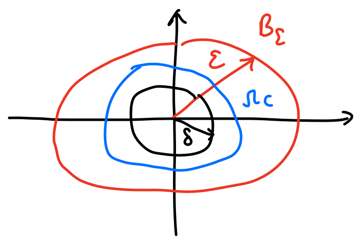
$\varepsilon_2 := \inf_{x \in \Omega_\varepsilon \setminus B_{\delta_1}} V(x) > 0$

Claim $\Omega_{\varepsilon_2/2}$ is bounded

To Prove: $x=0$ is stable

given any $\varepsilon > 0$, $\exists \, \delta(\varepsilon) > 0$

s.t. $|x_0| \le \delta \Rightarrow |x(t)| \le \varepsilon \quad \forall t \ge 0$



2 Proof by Contradiction

Suppose there is an initial condition x_0

s.t. starting from that I.C. $x(t) \not\to 0$

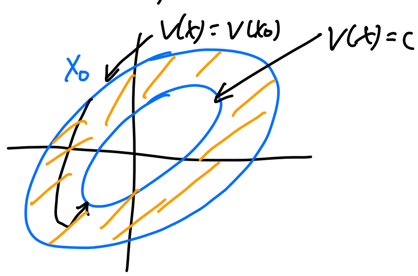
$$\Downarrow$$

$$V(x) \not\to 0$$

$c > 0$

But $V(x(t)) \longrightarrow c$

$(\dot{V} < 0 \text{ and } V \ge 0)$



$$\gamma = \max_{x:\, c \le V(x) \le V(x_0)} -\dot{V}(x)$$

$$\Rightarrow \quad \dot{V}(x) \le -\gamma \quad \forall t \ge 0$$

$$V(x) = V(x_0) + \int_0^t \dot{V}(x(\tau))\, d\tau$$

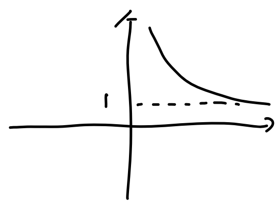
$$\le V(x_0) - \gamma t$$

< 0 for sufficiently high t

a contradiction to $V(x) \ge 0$

Therefore $x(t)$ will goes to 0

Ex. $f(x) = 1 + e^{-x}$



$$\frac{df}{dx} = -e^{-x} < 0$$

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3. If, in addition, $D = \mathbb{R}^n$ and $|x| \to \infty \Rightarrow V(x) \to \infty$ (radially unbounded)

then $x = 0$ is GAS

Ex. $V(x) = \frac{x_1^2}{1+x_1^2} + x_2^2$

Not radially unbounded

Ex. $\dot{x} = -g(x)$, $x \in \mathbb{R}$

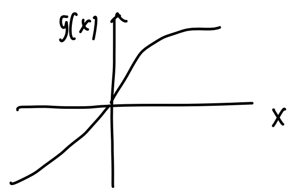
$V(x) = \frac{1}{2}x^2$

$\dot{V}(x) = \frac{\partial V}{\partial x}\dot{x}$

$= 2x(-g(x))$

< 0 if $x \neq 0$

$x = 0$ is GAS



$x\,g(x) > 0$

Ex. $\ddot{x} = -a\dot{x} - g(x)$

$x_1 = x$

$x_2 = \dot{x}_1$

$\dot{x}_2 = \ddot{x} = -a x_2 - g(x_1)$

$\dot{x}_1 = x_2$

$V(x) = \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2$

$\dot{V}(x) = \nabla V(x)^T f(x)$

$[x_1 \;\; x_2]\begin{bmatrix} x_2 \\ -a x_2 - g(x_1) \end{bmatrix}$

$\nabla V(x) = \begin{bmatrix} x_1 \end{bmatrix}$

$\lfloor x_2 \rfloor$

$$\dot V(x) = x_1 x_2 + x_2(-a x_2 - g(x_1))$$

$$= x_1 x_2 - a x_2^2 - x_2 g(x_1)$$

$$V(x) = \int_0^{x_1} g(y)\,dy + \frac{1}{2} x_2^2$$

$$\frac{\partial V}{\partial x_1} = g(x_1)$$

$$\frac{\partial V}{\partial x_2} = x_2$$

$$\dot V = g(x_1) x_2 + x_2(-a x_2 - g(x_1))$$

$$= g(x_1) x_2 - a x_2^2 - x_2 g(x_1)$$

$$\dot V = -a x_2^2$$

$a<0$: no conclusion

$a=0$: stable

$a>0$: stable

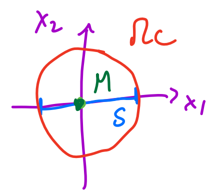
$x=0$ is stable

Lasalle: $\Omega_c \leftarrow$ disks

S: x_1 axis

$\Downarrow$

$x=0$ is GAS



Lasalle-Krasovskii Invariance Principle

- Time invariant systems
- Allow to conclude AS from $\dot V(x) \le 0$ if additional conditions hold